

Mortensen–Pissarides (1994)

A Readable Mathematical Rendering of the Lean 4 Formalization

Generated from `Lean4Tutorial.i002_replicate_MP`

20 July 2026

Purpose. This document translates the complete mathematical interface of the Lean 4 development into conventional notation. It is intended for manual comparison with Dale T. Mortensen and Christopher A. Pissarides, “Job Creation and Job Destruction in the Theory of Unemployment,” *Review of Economic Studies* 61 (1994), 397–415. It is not a replacement for the Lean kernel check; it is the human-readable companion to that check.

Lean build target	<code>lake build Lean4Tutorial.i002_replicate_MP</code>
Kernel-checked results	31 named theorems, each audited with <code>assert_no_sorry</code>
Mathematical source	<code>Definitions.lean</code> , <code>Equilibrium.lean</code> , and <code>Results.lean</code>
Verification source	<code>Verification.lean</code>
Document status	Built from the current Lean sources

Reading warning. A Lean *structure field* is assumed when an object of that structure is supplied. It is not automatically a derived theorem. This matters for the cyclical equilibrium structures, where some comparative-static orderings are encoded as fields. Section 9 separates assumptions, definitions, and derived results.

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1 How to read this audit

Every substantive formula below is paired with its Lean declaration. The translation uses ordinary mathematical notation while preserving the exact content of the Lean term. In particular:

- a **definition** introduces notation or a proposition;
- an **equilibrium structure** packages variables together with conditions that must be provided by any instance;
- a **theorem** is a kernel-checked implication from explicit hypotheses;
- an **assert_no_sorry** check rules out dependence on Lean’s **sorry** for the named theorem.

The Lean source uses P. «**lambda**» internally for the paper’s λ , because bare λ is reserved syntax in Lean. This document always prints the paper symbol λ .

1.1 Source files

File	Role
Definitions.lean	Primitives, matching technology, Bellman predicates, reduced equilibrium equations, gross flows, and period accounting.
Equilibrium.lean	Stationary value equilibrium, reduced steady state, two-state and Markov interfaces, and simulation paths.
Results.lean	All derived propositions currently proved by the development.
Verification.lean	One assert_no_sorry command for each of the 31 theorems in Results.lean .
i002_replicate_MP.lean	Dedicated build entry point importing the four modules above.

2 Primitives and maintained assumptions

Source: Definitions.lean: Primitives, Assumptions

An instance P of **Primitives** contains the following exogenous objects.

Symbol	Lean field	Meaning
r	r	Continuous-time discount rate.
λ	« lambda »	Poisson arrival rate of job-specific shocks.
σ	sigma	Dispersion of job-specific productivity.
β	beta	Worker’s Nash share of match surplus.
c	c	Flow cost of maintaining a vacancy.
b	b	Unemployment income or value of leisure.
p	p	Aggregate component of productivity.
ε_u	epsilon_u	Upper support point of the idiosyncratic shock.
dF	dF	Probability measure generating the shock distribution.
$q(\theta)$	q	Vacancy meeting rate.

The maintained **Assumptions P** proposition is exactly

$$\begin{aligned}
& r > 0, \quad \lambda \geq 0, \quad \sigma > 0, \quad 0 < \beta < 1, \quad c > 0, \\
& dF(\mathbb{R}) = 1, \quad dF((-\infty, \varepsilon_u]) = 1, \\
& 0 \leq F(x) \leq 1, \quad F \text{ is nondecreasing}, \\
& \theta > 0 \implies q(\theta) > 0, \quad q \text{ is nonincreasing}, \\
& \theta \longmapsto \theta q(\theta) \text{ is nondecreasing}.
\end{aligned}$$

The CDF bounds and monotonicity are fields, rather than results derived in Lean from the measure definition. This is a deliberate strengthening of the interface.

3 Basic mathematical definitions

Source: `Definitions.lean: namespace MP1994.Primitives`

3.1 Distribution, matching, and productivity

$$\begin{aligned}
F(x) &:= dF((-\infty, x]), \\
\theta(v, u) &:= \frac{v}{u}, \\
m(\theta, 1) &:= \theta q(\theta), \\
m(v, u) &:= v q(v/u), \\
y(\varepsilon) &:= p + \sigma \varepsilon.
\end{aligned}$$

The Lean names are respectively **F**, **theta**, **mWorker**, **m**, and **price**. Notice that the code’s aggregate matching definition is $vq(v/u)$, while the worker meeting rate is $\theta q(\theta)$.

3.2 Continuation values and separations

$$\begin{aligned}
I(\varepsilon_d) &:= \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx, & \text{optionValueIntegral,} \\
T(\ell, h) &:= \int_{\ell}^h [1 - F(x)] dx, & \text{tailIntegral,} \\
\mathcal{P}(S) &:= \int_{\mathbb{R}} \max\{S(x), 0\} dF(x), & \text{positiveContinuation,} \\
s(\varepsilon_d) &:= \lambda F(\varepsilon_d), & \text{destructionHazard.}
\end{aligned}$$

The first two integrals are Lebesgue interval integrals with respect to dx ; $\mathcal{P}(S)$ is an integral with respect to the shock measure dF .

3.3 Closed-form cutoff surplus

$$\begin{aligned}
 S_d(\varepsilon; \varepsilon_d) &:= \frac{\sigma(\varepsilon - \varepsilon_d)}{r + \lambda}, & \text{S_cutoff,} \\
 J_d(\varepsilon; \varepsilon_d) &:= (1 - \beta)S_d(\varepsilon; \varepsilon_d), & \text{J_cutoff,} \\
 [W - U]_d(\varepsilon; \varepsilon_d) &:= \beta S_d(\varepsilon; \varepsilon_d), & \text{WminusU_cutoff.}
 \end{aligned}$$

4 Stationary value equations

Source: `Definitions.lean` and `ValueEquilibrium in Equilibrium.lean`

The structure `ValueEquilibrium P` supplies $\theta, \varepsilon_d, V, U, J, W, w, S$, with $\theta > 0$ and $\varepsilon_d < \varepsilon_u$, and requires all conditions below.

Lean declaration: `Primitives.VacancyBellman -- paper equation (1)`

$$rV = -c + q(\theta)[J(\varepsilon_u) - V]. \quad (1)$$

Lean declaration: `Primitives.FreeEntry -- paper equation (2)`

$$rV = 0. \quad (2)$$

Lean declaration: `Primitives.SurplusIdentity -- paper equation (3)`

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U \quad \text{for every } \varepsilon. \quad (3)$$

Lean declaration: `Primitives.NashSharing -- paper equation (4)`

$$W(\varepsilon) - U = \beta S(\varepsilon) \quad \text{for every } \varepsilon. \quad (4)$$

Lean declaration: `Primitives.FilledJobBellman -- paper equation (5)`

$$rJ(\varepsilon) = p + \sigma\varepsilon - w(\varepsilon) + \lambda(1 - \beta)[\mathcal{P}(S) - S(\varepsilon)]. \quad (5)$$

Lean declaration: `Primitives.WorkerBellman -- paper equation (6)`

$$rW(\varepsilon) = w(\varepsilon) + \lambda\beta[\mathcal{P}(S) - S(\varepsilon)]. \quad (6)$$

Lean declaration: `Primitives.UnemployedBellman -- paper equation (7)`

$$rU = b + m(\theta, 1)[W(\varepsilon_u) - U]. \quad (7)$$

Lean declaration: `Primitives.SurplusBellman` -- paper equation (8)

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda\mathcal{P}(S) - \beta m(\theta, 1)S(\varepsilon_u). \quad (8)$$

The structure additionally requires

$$S(\varepsilon_d) = 0, \quad S(\varepsilon) \geq 0 \iff \varepsilon_d \leq \varepsilon.$$

These are `cutoff_zero` and `reservation_rule`.

5 Reduced stationary equilibrium

Source: `Definitions.lean` and `SteadyStateEquilibrium` in `Equilibrium.lean`

Lean declaration: `Primitives.JobDestructionCondition` -- paper equation (10)

$$p + \sigma\varepsilon_d = b + \frac{\beta c}{1 - \beta}\theta - \frac{\sigma\lambda}{r + \lambda} \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx. \quad (10)$$

Lean declaration: `Primitives.dispersionCutoffResponse` -- right side of paper equation (11)

The code defines the quantity described in its documentation as $\sigma \partial\varepsilon_d / \partial\sigma$ at fixed θ :

$$R_\sigma(\theta, \varepsilon_d) := \frac{(r + \lambda)/\sigma}{r + \lambda F(\varepsilon_d)} \left[p - b - \frac{\beta c}{1 - \beta} \theta \right]. \quad (11R)$$

Lean defines the right-hand side; it does not define a derivative operator or prove that an implicit derivative equals this expression.

Lean declaration: `Primitives.S_cutoff` -- paper equation (12)

$$S_d(\varepsilon; \varepsilon_d) = \frac{\sigma(\varepsilon - \varepsilon_d)}{r + \lambda}. \quad (12)$$

Lean declaration: `Primitives.JobCreationCondition` -- paper equation (13)

$$q(\theta) = \frac{c}{1 - \beta} \frac{r + \lambda}{\sigma(\varepsilon_u - \varepsilon_d)}. \quad (13)$$

Lean declaration: `Primitives.BeveridgeCondition` -- paper equation (14)

$$u = \frac{\lambda F(\varepsilon_d)}{\lambda F(\varepsilon_d) + m(\theta, 1)}. \quad (14)$$

Lean declaration: `Primitives.unemploymentDrift` -- paper equation (15)

$$\dot{u} = (1 - u)\lambda F(\varepsilon_d) - um(\theta, 1). \quad (15)$$

The structure `SteadyStateEquilibrium P` is a triple $(\theta, \varepsilon_d, u)$ satisfying (10), (13), and (14), together with $\theta > 0$, $\varepsilon_d < \varepsilon_u$, and $0 \leq u \leq 1$. Its derived notation is

$$v := \theta u, \quad n := 1 - u, \quad C := um(\theta, 1), \quad D := (1 - u)\lambda F(\varepsilon_d).$$

6 Anticipated aggregate states and Markov equations

Source: `Definitions.lean` and `Equilibrium.lean`

Stars denote boom values and unstarred quantities denote recession values.

Lean declaration: `Primitives.BoomDestructionCondition` -- paper equation (20)

$$\begin{aligned} p^* + \sigma \varepsilon_d^* = & b + \frac{\beta c}{1 - \beta} \theta^* - \frac{\lambda \sigma}{r + \lambda + \mu} \int_{\varepsilon_d^*}^{\varepsilon_d} [1 - F(x)] dx \\ & - \frac{\lambda \sigma}{r + \lambda} \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx. \end{aligned} \quad (20)$$

Lean declaration: `Primitives.RecessionDestructionCondition` -- paper equation (24)

$$\begin{aligned} p + \sigma \varepsilon_d = & b + \frac{\beta c}{1 - \beta} \theta - \frac{\lambda \sigma}{r + \lambda} \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx \\ & - \frac{\mu \sigma}{r + \lambda + \mu} (\varepsilon_d - \varepsilon_d^*). \end{aligned} \quad (24)$$

Lean declaration: `Primitives.boomSurplus` -- paper equation (29)

$$S^*(\varepsilon) = \frac{\sigma(\varepsilon - \varepsilon_d^*)}{r + \lambda + \mu} + \frac{\mu}{r + \lambda + \mu} S_d(\varepsilon; \varepsilon_d). \quad (29)$$

Lean declaration: `Primitives.BoomJobCreationCondition` -- paper equation (30)

$$q(\theta^*) = \frac{c(r + \lambda)/(1 - \beta)}{\sigma(\varepsilon_u - \varepsilon_d^*) - \frac{\mu \sigma(\varepsilon_d - \varepsilon_d^*)}{r + \lambda + \mu}}. \quad (30)$$

6.1 Two-state structure fields

`AnticipatedTwoStateEquilibrium P` packages $(p, p^*, \mu, \theta, \theta^*, \varepsilon_d, \varepsilon_d^*)$ and requires

$$p < p^*, \quad \mu \geq 0, \quad \theta > 0, \quad \theta^* > 0, \quad \varepsilon_d^* < \varepsilon_d < \varepsilon_u,$$

equations (20), (24), (13) for recession, and (30) for the boom.

The simpler `TwoStateEquilibrium` `recession` `boom` packages two reduced steady states and directly requires

$$p_{\text{rec}} < p_{\text{boom}}, \quad \theta_{\text{rec}} < \theta_{\text{boom}}, \quad \varepsilon_{d,\text{boom}} < \varepsilon_{d,\text{rec}}.$$

Thus these three orderings are data of that structure, not the conclusions of a constructor theorem in the current development.

6.2 General finite-state value equilibrium

For a finite state type, `GeneralMarkovValueEquilibrium` provides $\theta_s, \varepsilon_{d,s}, S(\varepsilon, s), \mu_s$, and a transition matrix $G(t \mid s)$, with positive tightness, admissible cutoffs, nonnegative transition objects, and $\sum_t G(t \mid s) = 1$.

Lean declaration: `GeneralMarkovValueEquilibrium.free_entry` -- paper equation (32)

$$m(\theta_s, 1)(1 - \beta_s)S(\varepsilon_{u,s}, s) = c_s. \quad (32)$$

Lean declaration: `GeneralMarkovValueEquilibrium.cutoff_zero` -- paper equation (33)

$$S(\varepsilon_{d,s}, s) = 0. \quad (33)$$

Lean declaration: `GeneralMarkovValueEquilibrium.surplus_bellman` -- paper equation (34)

$$\begin{aligned} (r_s + \lambda_s + \mu_s)S(\varepsilon, s) &= p_s + \sigma_s \varepsilon - b_s - \beta_s m(\theta_s, 1)S(\varepsilon_{u,s}, s) \\ &\quad + \lambda_s \int \max\{S(x, s), 0\} dF_s(x) \\ &\quad + \mu_s \sum_t G(t \mid s) \max\{S(\varepsilon, t), 0\}. \end{aligned} \quad (34)$$

The separate `MarkovEquilibrium` structure is weaker: it contains a reduced steady-state equilibrium in each state and a row-stochastic G , but does not itself impose equations (32)–(34).

7 Simulation and accounting definitions

Source: `Definitions.lean` and `SimulationPath` in `Equilibrium.lean`

Lean declaration: `Primitives.periodCreation` -- paper equation (36)

$$C_t := a_t(1 - N_t). \quad (36)$$

Lean declaration: `Primitives.immediateScrapping` and `periodDestruction` -- paper equation (37)

$$D_t^{\text{immediate}} := \int_{\varepsilon_d^{\text{old}}}^{\varepsilon_d^{\text{new}}} n(x) dx, \quad D_t := D_t^{\text{immediate}} + D_t^{\text{ongoing}}. \quad (37)$$

Lean declaration: `Primitives.nextEmployment` -- paper equation (38)

$$N_{t+1} := N_t + C_t - D_t. \quad (38)$$

`SimulationPath P` contains sequences $N_t, a_t, D_t^{\text{immediate}}, D_t^{\text{ongoing}}, C_t, D_t$ and requires (36)–(38) at every natural-numbered period. The structure does not impose a particular formula on D_t^{ongoing} ; that sequence is supplied by an instance.

8 Kernel-checked theorem catalogue

This section translates every theorem named in `Results.lean`. Each row is also checked by an `assert_no_sorry` command in `Verification.lean`.

8.1 Value equations and cutoff algebra

Lean theorem	Exact mathematical content
<code>surplus_bellman</code>	Every <code>ValueEquilibrium</code> satisfies (8). The proof adds (5) and (6), subtracts (7), and uses (3)–(4).
<code>J_cutoff_eq_zero</code>	Every <code>ValueEquilibrium</code> satisfies $J(\varepsilon_d) = 0$, using $S(\varepsilon_d) = 0$, (3), and (4).
<code>vacancy_free_entry_value</code>	Under $r > 0$, equations (1)–(2) imply $q(\theta)J(\varepsilon_u) = c$.
<code>S_cutoff_self</code>	$S_d(\varepsilon_d; \varepsilon_d) = 0$.
<code>surplus_difference</code>	$S_d(\varepsilon_2; \varepsilon_d) - S_d(\varepsilon_1; \varepsilon_d) = \sigma(\varepsilon_2 - \varepsilon_1)/(r + \lambda)$.
<code>nash_shares_exhaust_surplus</code>	$J_d(\varepsilon; \varepsilon_d) + [W - U]_d(\varepsilon; \varepsilon_d) = S_d(\varepsilon; \varepsilon_d)$.
<code>surplus_nonneg</code>	Under the maintained sign assumptions, $\varepsilon_d \leq \varepsilon \Rightarrow S_d(\varepsilon; \varepsilon_d) \geq 0$.
<code>J_cutoff_nonneg</code>	Under the same assumptions, $\varepsilon_d \leq \varepsilon \Rightarrow J_d(\varepsilon; \varepsilon_d) \geq 0$.

8.2 Matching, job creation, and destruction

Lean theorem	Exact mathematical content
<code>destructionHazard_nonneg</code>	$\lambda F(\varepsilon_d) \geq 0$.
<code>destructionHazard_mono</code>	$\varepsilon_d \mapsto \lambda F(\varepsilon_d)$ is nondecreasing.
<code>mWorker_pos</code>	$\theta > 0 \Rightarrow m(\theta, 1) > 0$.
<code>jobCreation_iff_freeEntryValue</code>	If $\varepsilon_d < \varepsilon_u$, equation (13) is equivalent to $q(\theta)J_d(\varepsilon_u; \varepsilon_d) = c$.

Lean theorem	Exact mathematical content
jobCreationCurve_slopes_down	For two points satisfying (13), if $\varepsilon_1 < \varepsilon_2 < \varepsilon_u$, then $\theta_2 < \theta_1$ (using that q is nonincreasing).
dispersionCutoffResponse_pos	If $b + [\beta c / (1 - \beta)]\theta < p$, then $R_\sigma(\theta, \varepsilon_d) > 0$. This proves the sign of the defined right-hand side (11R), not an implicit-function theorem.
boomJobCreation_zero_transition_iff	Setting $\mu = 0$ in (30) is equivalent to (13), with the boom cutoff.
anticipation_reduces_boom_tightness	Fix cutoffs with $\varepsilon_d^* < \varepsilon_d < \varepsilon_u$ and $\mu > 0$. If the denominator of (30) is positive, a no-transition point satisfying (13) and an anticipated point satisfying (30) obey $\theta_\mu^* < \theta_0^*$.

8.3 Flows and stationary accounting

Lean theorem	Exact mathematical content
beveridge_denominator_pos	If $\theta > 0$, then $\lambda F(\varepsilon_d) + m(\theta, 1) > 0$.
creationFlow_mono	Holding $u \geq 0$ fixed, creation is nondecreasing in the worker meeting rate.
destructionFlow_mono	Holding $u \leq 1$ fixed, destruction is nondecreasing in ε_d .
aggregateShock_flow_orders	Conditional on $\theta_0 \leq \theta_1$ and $\varepsilon_{d,1} \leq \varepsilon_{d,0}$, creation weakly rises and destruction weakly falls. The equilibrium ordering is a hypothesis.
dispersionShock_flow_orders	Conditional on $\theta_0 \leq \theta_1$ and $\varepsilon_{d,0} \leq \varepsilon_{d,1}$, creation and destruction both weakly rise. The equilibrium ordering is a hypothesis.
gross_flows_balance	Every <code>SteadyStateEquilibrium</code> satisfying the maintained assumptions has $C = D$.
unemploymentDrift_eq_zero	Every such steady state has $\dot{u} = 0$.
v_eq_theta_mul_u	The equilibrium vacancy definition gives $v = \theta u$.

8.4 Qualitative co-movement and period accounting

Lean defines

$$\text{MovesOpposite}(x_0, x_1, y_0, y_1) : \Longleftrightarrow (x_1 - x_0)(y_1 - y_0) < 0,$$

$$\text{MovesTogether}(x_0, x_1, y_0, y_1) : \Longleftrightarrow 0 < (x_1 - x_0)(y_1 - y_0).$$

Lean theorem	Exact mathematical content
aggregateShock_negative_comovement	From strict hypotheses $C_0 < C_1$ and $D_1 < D_0$, it follows that creation and destruction satisfy <code>MovesOpposite</code> .
dispersionShock_positive_comovement	From strict hypotheses $C_0 < C_1$ and $D_0 < D_1$, it follows that the flows satisfy <code>MovesTogether</code> .

Lean theorem	Exact mathematical content
<code>unemploymentDrift_pos_iff</code>	$\dot{u} > 0 \iff C < D.$
<code>employment_change_identity</code>	$(N + C - D) - N = C - D.$
<code>downturn_destruction_exceeds_ongoing</code>	If $D^{\text{immediate}} > 0$, then $D^{\text{ongoing}} < D^{\text{immediate}} + D^{\text{ongoing}}.$
<code>boom_without_scrapping</code>	$0 + D^{\text{ongoing}} = D^{\text{ongoing}}.$
<code>SimulationPath.employment_change</code>	Every simulation path satisfies $N_{t+1} - N_t = C_t - D_t.$

9 What is proved, assumed, or only represented

Status	Topic	Current formal content
PROVED	Value-equation aggregation	Equations (3)–(7) imply (8); free entry implies the zero-profit value equation; cutoff firm value is zero.
PROVED	Stationary accounting	The Beveridge equation implies $C = D$ and $\dot{u} = 0$; period employment changes by $C - D$.
PROVED	Conditional monotonicity	Hazard, creation, destruction, JC-locus, and anticipation statements follow from the explicitly listed sign and ordering hypotheses.
ASSUMED	Aggregate-shock equilibrium ordering	The rise in θ and fall in ε_d are fields or hypotheses before the flow-order and co-movement lemmas are applied.
ASSUMED	Dispersion-shock equilibrium ordering	The joint rise in θ and ε_d is a hypothesis of the corresponding flow theorem. The code proves positivity of (11R) under a net-price condition, but not the full joint implicit comparative static.
REPRESENTED	Markov equilibrium	Equations (32)–(34) are encoded as fields of a value-equilibrium structure; no existence, uniqueness, or solution theorem is currently proved.
REPRESENTED	Simulation	Equations (36)–(38) and the immediate scrapping integral are encoded. Calibration, numerical solution, and the paper’s simulated series are not reproduced.
OUT OF SCOPE	Full paper replication	The current Lean development does not formalize every equation, derive all cyclical orderings from primitives, prove equilibrium existence/uniqueness, or replicate the Section 5 calibration.

This classification follows the actual types in the Lean files. It is the key manual audit: an implication whose economic conclusion is already a hypothesis is logically valid but does not by itself reproduce the paper’s derivation of that hypothesis.

10 Verification ledger

10.1 No-sorry audit

`Verification.lean` imports `Results.lean` and applies `assert_no_sorry` to all 31 declarations listed in Section 8. The command asks Lean to collect the axioms used by each declaration and fails if `sorryAx` occurs. Therefore a successful dedicated build establishes both elaboration and absence of unfinished `sorry` proofs for this theorem set.

10.2 Reproduction commands

From the repository root:

```
lake build Lean4Tutorial.i002_replicate_MP
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -output-directory=output/pdf \
  output/pdf/mp1994_lean_formalization.tex
```

10.3 Manual comparison checklist

1. Compare the primitive definitions and signs in Sections 2–3 with the paper.
2. Compare equations (1)–(15), (20), (24), (29)–(30), (32)–(34), and (36)–(38), paying special attention to integration measures and denominators.
3. Check whether the paper’s $m(\theta, 1)$ convention matches $\theta q(\theta)$ throughout.
4. Distinguish theorems that derive an ordering from those that accept it as a hypothesis or structure field.
5. Decide whether the omitted existence, uniqueness, implicit-function, and calibration arguments are required for the intended meaning of “replication.”

Reference

Dale T. Mortensen and Christopher A. Pissarides (1994), “Job Creation and Job Destruction in the Theory of Unemployment,” *Review of Economic Studies*, 61(3), 397–415. The locally supplied paper is `Mortensen-JobCreationJob-1994.pdf`.